Project: Customer Churn Analysis for Telecommunications Company
Team: Team 3
Stage: Stage 2 – Data Preparation and Clustering Analysis
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Role: Project manager

1. Purpose

Feature scaling was applied during the data preparation stage to ensure that numerical features with different ranges were represented on a comparable scale before predictive modelling and clustering analysis.

The dataset contains numerical variables such as `tenure` and `MonthlyCharges`. Because these variables have different ranges, leaving them unscaled could cause features with larger numerical values to have a greater influence on the analysis, particularly for distance-based techniques such as K-Means clustering.

For this project, **StandardScaler** from the Scikit-learn library was used to standardise the numerical features.

2. Scaling Technique Used

The scaling technique selected for the project was **StandardScaler**.

StandardScaler transforms numerical values by subtracting the mean of the feature and dividing by its standard deviation. This results in transformed features with a mean approximately equal to **0** and a standard deviation approximately equal to **1**.

The following numerical features were scaled:

- `tenure`
- `MonthlyCharges`

The remaining features were binary or encoded categorical variables and were not included in the StandardScaler transformation.

3. Training and Testing Dataset

After data cleaning and preprocessing, the prepared dataset contained **6,741 customer records**.

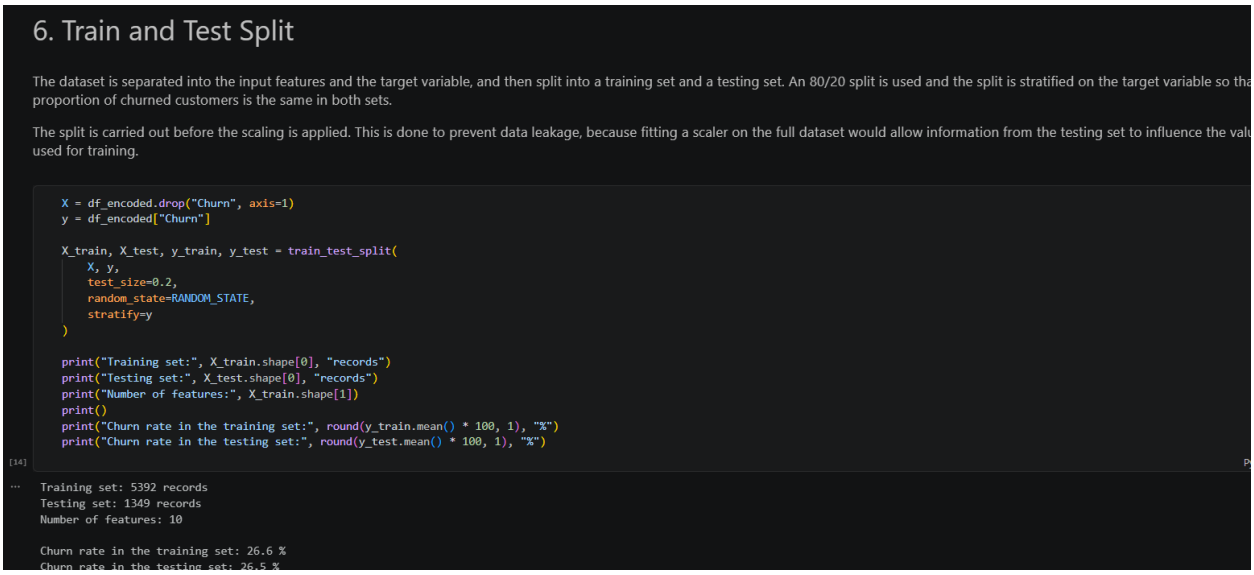
For predictive modelling, the dataset was divided into separate training and testing sets using an **80/20 split**.

The resulting datasets contained:

- **Training set:** 5,392 customer records
- **Testing set:** 1,349 customer records
- **Number of predictive features:** 10
- **Target variable:** `Churn`

A `random_state` value of **42** was used to make the split reproducible. The `stratify=y` parameter was also applied so that the distribution of the Churn target remained similar between the training and testing datasets.

The following code was used:

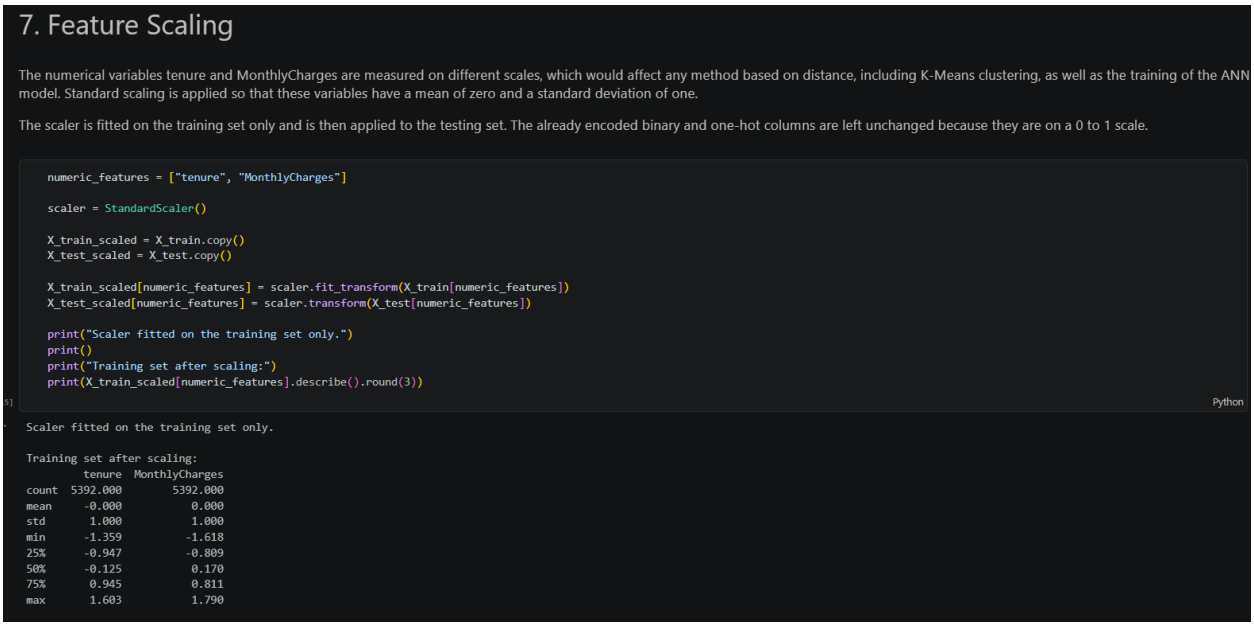


4. Scaling the Training and Testing Data

For predictive modelling, StandardScaler was fitted using **only the training dataset**.

The fitted scaler was then used to transform both the training and testing datasets. Fitting the scaler only on the training data is important because it prevents information from the testing dataset from influencing the preprocessing of the training data. This helps avoid **data leakage** and provides a more reliable basis for model validation.

The following code was used:



After scaling, the training values for `tenure` and `MonthlyCharges` had means close to **0** and standard deviations close to **1**, confirming that the StandardScaler transformation had been applied successfully.

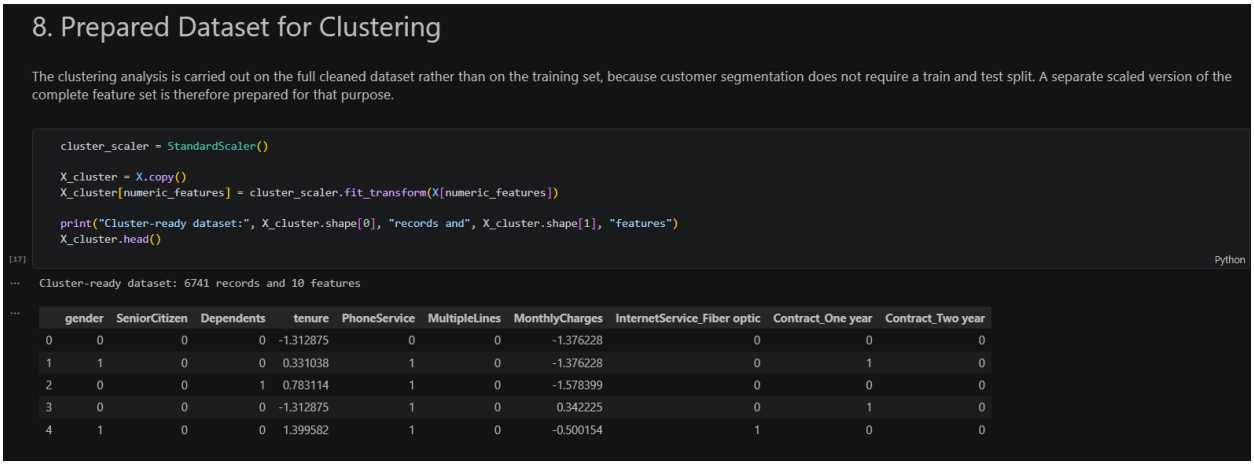
5. Scaling for K-Means Clustering

A separate scaling process was used to prepare the dataset for the clustering analysis.

K-Means clustering uses distances between observations when assigning customers to clusters. Therefore, scaling the numerical variables was important to prevent numerical features with larger ranges from having disproportionate influence on the clustering process.

For clustering, the complete feature dataset was copied and StandardScaler was applied to `tenure` and `MonthlyCharges`.

The following code was used:



8. Prepared Dataset for Clustering

The clustering analysis is carried out on the full cleaned dataset rather than on the training set, because customer segmentation does not require a train and test split. A separate scaled version of the complete feature set is therefore prepared for that purpose.

```
cluster_scaler = StandardScaler()

X_cluster = X.copy()
X_cluster[numeric_features] = cluster_scaler.fit_transform(X[numeric_features])

print("Cluster-ready dataset:", X_cluster.shape[0], "records and", X_cluster.shape[1], "features")
X_cluster.head()
```

[17] Python

```
Cluster-ready dataset: 6741 records and 10 features
```

	gender	SeniorCitizen	Dependents	tenure	PhoneService	MultipleLines	MonthlyCharges	InternetService_Fiber optic	Contract_One year	Contract_Two year
0	0	0	0	-1.312875	0	0	-1.376228	0	0	0
1	1	0	0	0.331038	1	0	-1.376228	0	1	0
2	0	0	1	0.783114	1	0	-1.578399	0	0	0
3	0	0	0	-1.312875	1	0	0.342225	0	1	0
4	1	0	0	1.399582	1	0	-0.500154	1	0	0

The resulting clustering-ready dataset contained:

- **6,741 customer records**
- **10 clustering features**
- **0 missing values**

The target variable Churn was **not included as an input feature for K-Means clustering**, because clustering is an unsupervised learning technique and the objective was to identify customer segments based on customer characteristics rather than the known churn outcome.

The clustering-ready dataset was exported as:

`ATS_cluster_ready_v1.csv`

6. Prepared Output Files

The data preparation process produced separate files for use in subsequent analysis and modelling:

- `ATS_clean_v1.csv` – cleaned and encoded customer dataset.
- `ATS_cluster_ready_v1.csv` – scaled dataset prepared for K-Means clustering.
- `ATS_train_features_v1.csv` – training feature dataset.

- `ATS_test_features_v1.csv` – testing feature dataset.
- `ATS_train_target_v1.csv` – training target values.
- `ATS_test_target_v1.csv` – testing target values.

These files allow the predictive modelling and clustering stages to use consistent and reproducible prepared data.

7. Summary

StandardScaler was successfully applied to the numerical features `tenure` and `MonthlyCharges` as part of the Stage 2 data preparation process.

For predictive modelling, the scaler was fitted only on the training dataset and subsequently applied to the testing dataset to minimise the risk of data leakage. For K-Means clustering, a separate scaler was fitted to the complete feature dataset to prepare all 6,741 customer records for customer segmentation.

The scaling process ensures that numerical features are represented on comparable scales and prepares the dataset for subsequent predictive modelling and clustering analysis.